Week-6

1. According to the lecture, why would standard cryptographic solutions be unsuitable for AI model privacy? **(Select all that apply.)**
    a. It is unclear whether all complex functions can be effectively evaluated from in the encryption space.
    b. Evaluation in encrypted space is computationally expensive.
    c. As the model scales, cryptographic keys become impossible to manage.
    d. Encryption models cannot be used for binary classification tasks.

    Explanation: Data Privacy [4:06-6:22] - There are two reasons why cryptographic solutions are not suitable. First, it is unclear whether you can do inference in an encrypted space. Second, encryption is an expensive solution.

2. In the context of healthcare data, why might a trained binary classification model still pose a privacy risk even if the raw data is not directly released?
    a. The model weights occupy too much storage space.
    b. Binary classifiers are inherently less secure than multi-class ones.
    c. An adversary can reverse-engineer sensitive training data.
    d. The labels used for training are usually public knowledge.

    Explanation: Data Privacy [1:45-4:06] - An adversary can reverse-engineer sensitive personal information such as healthcare or financial data given a simple binary classification model.

3. How were users able to deanonymize the “anonymized” Netflix dataset?
    a. By obtaining the meta-data of the dataset.
    b. By brute-forcing the sparse matrix to reveal the missing entries.
    c. By using cryptanalytic attacks to decode user information.
    d. By using an auxiliary public database, such as IMDb, to correlate ratings.

    Explanation: Data Privacy [8:55-12:46] - They were able to deanonymize the Netflix dataset by using an auxiliary database (IMDb) and correlating the two.

4. In the randomized response algorithm presented in the lecture, what is the minimum probability that the truthful answer is revealed?
    a. 10%
    b. 25%
    c. 50%
    d. 75%

    Explanation: Data Privacy [21:25-24:12] - $\varepsilon > 0$ means that the truth will be revealed with a probability of $\frac{e^{\varepsilon}}{1+e^{\varepsilon}} > 50\%$.

5. To maximize utility, how should the privacy parameter ε be adjusted?
    a. ε should be maximum for privacy, ε should be minimum for utility
    b. ε should be minimum for privacy, ε should be minimum for utility

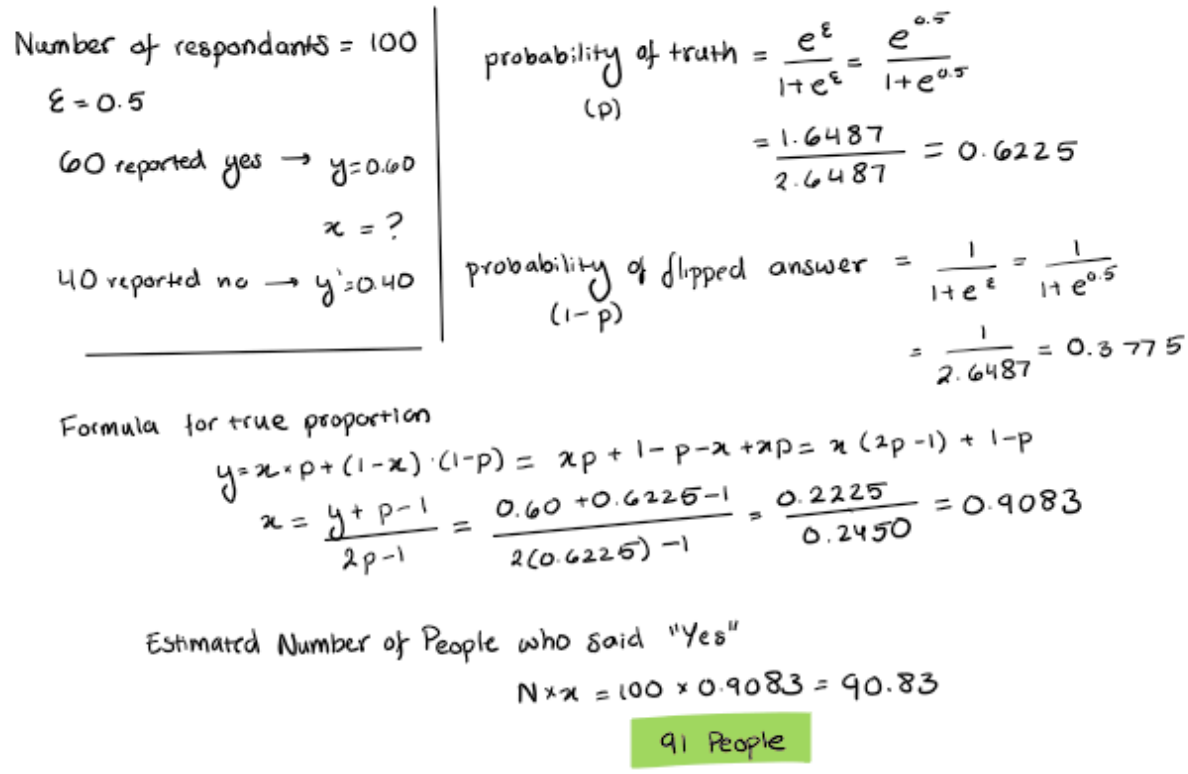
c. ε should be maximum for privacy, ε should be maximum for utility
d. ε should be minimum for privacy, ε should be maximum for utility
e. ε is unrelated

Explanation: Data Privacy [1:00:00 - 1:01:20] - If ε is large, x_i and z_i are easier to distinguish, which increases utility, and vice versa. ε controls the privacy-utility tradeoff.

6. An organization surveys 100 people and asks whether they voted for a certain party. To protect respondents' privacy, the organization uses the randomized response technique with $\varepsilon = 0.5$. Out of 100 respondents, 60 reported “Yes” and 40 reported “No.” Estimate the true number of people whose actual answer was “Yes”. Round to the nearest whole number.

Answer (numerical): 91

Explanation: Data Privacy



7. Which trust scenario correctly describes the differential privacy model?
   a. Trust the curator; Trust the world
   b. Do not trust the curator; Trust the world
   c. Trust the curator; Do not trust the world
   d. Do not trust the curator; Do not trust the world

Explanation: Data Privacy [1:06:06-1:11:00] - In differential privacy, surveyors trust the curator to give their information to them but want their information to be privatized when the computed function is released to the world.

8. In the Laplace Mechanism, how is the sensitivity (Δ) defined for a function f?
    a. The average difference between $f(X)$ and $f(X')$ for all neighbouring X and X'.
    b. The maximum absolute difference between $f(X)$ and $f(X')$ for any two neighboring X and X'.
    c. The standard deviation of the function f.
    d. The minimum difference between $f(X)$ and $f(X')$ for any two neighboring X and X'.

    Explanation: Differential Privacy [21:30-22:34] - Sensitivity (Δ) in the Laplace mechanism measures how sensitive a function f is to a single input change. It is the maximum absolute difference $|f(X) - f(X')|$ over all neighboring datasets X and X' that differ by one element.

9. According to Cynthia Dwork's definition of ε-differential privacy, for all neighbouring datasets X and X', the probability that a mechanism M on X generates an output in subset S divided by the probability that a mechanism M on X' generates an output in subset S is at most:
    a. e^{ε}
    b. ε
    c. $1/\varepsilon$
    d. ε^{e}

    Explanation: Differential Privacy [9:57-15:02] - A mechanism M is said to be ε -differential private if $\frac{Pr(M(X) \in S)}{Pr(M(X') \in S)} \leq e^{\varepsilon}$.

10. What property of neighboring data sets X and X' is used to prove the privacy guarantee of Randomized Response?
    a. They have zero correlation.
    b. They differ by 99% of the elements in the set.
    c. They are disjoint sets.
    d. They differ by only one element.

    Explanation: Data Privacy [24:29-25:58] - X and X' are two datasets where $X = X' \forall i \in \{1 \dots (n-1)\}$, meaning they differ by only one element of index $i = n$.

11. For a fixed ε, if a function being computed is highly sensitive (large Δ), how does it affect the noise added to the Laplace mechanism?
    a. Less noise is added.
    b. More noise is added.
    c. The noise remains unchanged.
    d. Noise is irrelevant.

    Explanation: Differential Privacy [27:14-31:00] - The more sensitive the function, the more noise should be added.

12. Comparing the Laplace mechanism and randomized response, how does the required number of respondents n differ to achieve the same privacy–utility trade-off?
    a. The Laplace mechanism requires quadratically more people.
    b. The Laplace mechanism requires quadratically fewer people.
    c. Both require the same number of people.
    d. The number of people to survey does not matter.

    Explanation: Differential Privacy [50:18-53:02] - If you trust the curator, then you would have to survey a quadratically lesser number of people to achieve the same privacy-utility tradeoff using the Laplace mechanism than not trusting the curator.